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Statistics homework
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Irrationality, Normality, Transcendence

(and the Number of the Beast)

I - Digit density of some irrational numbers.

The transcendental number π is suspected to be a *normal number*, i.e. a number that has an equal frequency for each of its digits (and beyond, an equal density of any sequence of n digits, for all n). Let's try to verify this property over a (necessarily) small sample of digits.

1) Compute the frequency f_d of each of the ten digits $d \in [0, 9]$ in a sequence of $N = 5 \times 10^4$ digits of π (standard base-10) [1].
2) Draw the histogram of frequencies, f_d versus d, with error-bars[2] $\pm\sigma_{f_d}$.
3) We want to examine the equiprobability hypothesis. Using the *scipy.optimize.curve_fit()* method, fit the two-parameter linear function:

$$f^{lin}(d; \alpha, \beta) = \alpha/10 + \beta \times d$$

to the histogram data points with the least-squares method. The equiprobability hypothesis $f_d = 1/10$ would correspond to the parameters point $\vec{\mu}_0 = (\alpha_0, \beta_0) = (1.0, 0.0)$

4) Give the best-fit parameters $\vec{\hat{\mu}}_p = (\hat{\alpha}, \hat{\beta})$ with their associated uncertainty and give their correlation[3]. Compute the $\chi^2_{min} = \chi^2(\hat{\alpha}, \hat{\beta})$ value:

$$\chi^2_{min} = \chi^2(\hat{\alpha}, \hat{\beta}) = \sum_{d=0}^{9} \left(\frac{f_d - f^{lin}(d; \hat{\alpha}, \hat{\beta})}{\sigma_{f_d}} \right)^2,$$

that quantifies the agreement of the linear hypothesis with the data points (the larger the χ^2_{min}, the smaller the agreement). If the statistical fluctuations around the theoretical model are Normally-distributed we expect the χ^2_{min} per degree-of-freedom to be close to unity. Evaluate the corresponding *p-value*, defined as the survival probability of the χ^2-distribution above the observed χ^2_{min} value for $n_{dof} = n_{points} - n_{params} = 10 - 2 = 8$.

5) Using the covariance C returned by the least-squares fit, draw the covariance error ellipse corresponding to a 95% confidence level in the two-dimensional parameter space $\vec{p} = (\alpha, \beta)$:

$$\Delta\chi^2(\alpha, \beta) = \chi^2(\alpha, \beta) - \chi^2_{min} = {}^{t}(\vec{p} - \vec{\hat{\mu}}_p) C^{-1} (\vec{p} - \vec{\hat{\mu}}_p) = \text{chi2.ppf}(0.95, 2).$$

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[1] You may use for instance the *mpmath* python library that provides floating-point arithmetic with arbitrary precision. E.g.:

```
from mpmath import mp
mp.dps = N    # set number of digits
num=str(mp.pi)
```

then scan the *num* string (given in appendix). Adapt the value of N to your computational ressources.

[2] The content of the histogram bins being large, we can assume a symmetrical uncertainty derived from the binomial variance:

$$\mathbb{V}[f_d] = \sigma^2_{f_d} = f_d(1 - f_d)/N$$

[3] The *curve_fit()* method returns the covariance matrix of the parameters

Indicate on the figure the best-fit point $\vec{\hat{\mu}}_p = (\hat{\alpha}, \hat{\beta})$ (i.e. the ellipse centre, $\Delta\chi^2 = 0$) and the equiprobability hypothesis, $\vec{\mu}_0 = (\alpha_0, \beta_0)$. Evaluate the $\Delta\chi^2(\alpha_0, \beta_0)$ value and give the corresponding *p-value*: $p = \text{chi2.sf}\,(\Delta\chi^2(\alpha_0, \beta_0), 2)$, that quantifies the consistancy of the equal-probability hypothesis with the best-fit linear model. Conclusions [4]

6) Replay the same game with few other well-known irrational numbers, for instance, the Euler constant e (*mpmath.euler*) or the algebric numbers: $\sqrt{2}$ (*mpmath.sqrt{2}*) and the golden ratio $\varphi = (1+\sqrt{5})/2$ (*mpmath.phi*).
   Another interesting (and related) digits sequence you should easily test, is the concatenation of the Fibonacci numbers ($F_0 = F_1 = 1$; $F_n = F_{n-1} + F_{n-2}$; related to the golden ratio[5] as $\lim_{n\to\infty}(F_{n+1}/F_n) = \varphi$). Other similar increasing sequences do exhibit a similar "equal-probability" behaviour, e.g.: the Lucas sequence ($L_0 = 2$; $L_1 = 1$; $L_n = L_{n-1} + L_{n-2}$; $\lim_{n\to\infty}(L_n/F_n) = \sqrt{5}$) or the Padovan sequence ($P_0 = P_1 = P_2 = 1$, $P_n = P_{n-2} + P_{n-3}$; related to the algebric *plastic number*: $\lim_{n\to\infty}(P_{n+1}/P_n) = \rho = \sqrt[3]{\frac{9+\sqrt{69}}{18}} + \sqrt[3]{\frac{9-\sqrt{69}}{18}} \simeq 1.32471...$).

II - The law of *anomalous* numbers.

The first (non-zero) numerical digit of any random numbers you meet in everyday life (reading newspaper, sport results, measurements, ...) are not equally distributed. The digit $d = 1$ is over-represented as first-digit, with a frequency exceeding 30%, while the rate of poor digit $d = 9$ is below 5%, far from the equiprobability hypothesis $f_d = 1/9$. The first-digit frequency is well described by the discrete *Newcomb-Benford* probability mass function (base-10):

$$f_d^B = log_{10}(d+1) - log_{10}(d) \quad (d \in [1,9]) \;\; ; \;\; \sum_{d=1}^{9} f_d^B = log_{10}(10) - log_{10}(1) = 1$$

In particular, the 1st-digit of the Fibonacci numbers, F_n closely follow this frequency distribution.

1) Following the steps of section I, extract the frequencies f_d ($d \in [1,9]$) in the 1st-digit of the Fibonacci numbers F_n ($n = 0, N$); draw the frequency histogram with error-bars and adjust the ad-hoc model :
   $$f^{log}(d) = log_{10}(d+\alpha) - log_{10}(d+\beta)$$
   to the data points. The exact Benford-distribution hypothesis would correspond to the parameters point $\vec{\mu}_0 = (\alpha_0, \beta_0) = (1.0, 0.0)$. Draw the 95% confidence ellipse around the best-fit parameters $\vec{\hat{\mu}} = (\hat{\alpha}, \hat{\beta})$ and quantify the agreement with the exact Benford hypothesis.
2) It has been shown (Boyle, 1994) that any number that results from the multiplication of a large number of independent random variables asymptotically follow the Benford law. Following the above steps, check this behaviour for the variable $Z_n = \prod_{k=0}^{n} X_k$, where X_k are uniformly-distributed random variables, for different increasing values of n.
3) Similarly, check that the 1st-digit of the variable $Z_n = (X/Y)^n$, where both X and Y are uniformly-distributed, rapidly tends to the Bendford law for $n \gtrsim 5$.

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[4] This is not a formal proof of normality. Such proof is still to be demonstrated.

[5] The omnipresent golden ratio is also related to the scary *Number of the Beast*: $sin(666) = cos(6 \times 6 \times 6) = -\varphi/2$.

π=3.1415926535897932384626433832795028841971693993751058209749445923078164062862089986280348253421170679821480865132823066470938446095505822317253594081284811174502841027019385211055596446229489549303819644288109756659334461284756482337867831652712019091456485669234603486104543266482133936072602491412737245870066063155881748815209209628292540917153643678925903600113305305488204665213841469519415116094330572703657595919530921861173819326117931051185480744623799627495673518857527248912279381830119491298336733624406566430860213949463952247371907021798609437027705392171762931767523846748184676694051320005681271452635608277857713427577896091736371787214684409012249534301465495853710507922796892589235420199561121290219608640344181598136297747713099605187072113499999983729780499510597317328160963185950244594553469083026425223082533446850352619311881710100031378387528865875332083814206171776691473035982534904287554687311595628638823537875937519577818577805321712268066130019278766111959092164201989380952572010654858632788659361533818279682303019520353018529689957736225994138912497217752834791315155748572424541506959508295331168617278558890750983817546374649393192550604009277016711390098488240128583616035637076601047101819429555961989467678374494482553797747268471040475346462080466842590694912933136770289891521047521620569660240580381501935112533824300355876402474964732639141992726042699227967823547816360093417216412199245863150302861829745557067498385054945885869269956909272107975093029553211653449872027559602364806654991198818347977535663698074265425278625518184175746728909777727938000816470600161452491921732172147723501414419735685481613611573525521334757418494684385233239073941433345477624168625189835694855620992192221842725502542568876717904946016534668049886272327917860857843838279679766814541009538837863609506800642251252051173929848960841284886269456042419652850222106611863067442786220391949450471237137869609563643719172874677646575739624138908658326459958133904780275900994657640789512694683983525957098258226205224894077267194782684826014769909026401363944374553050682034962524517493996514314298091906592509372216964615157098583874105978859597729754989301617539284681382686838689427741559918559252459539594310499725246808459872736446958486538367362226260991246080512438843904512441365497627807977156914359977001296160894416948685558484063534220722258284886481584560285060168427394522674676788952521385225499546667278239864565961163548862305774564980355936345681743241125150760694794510965960940252288797108931456691368672287489405601015033086179286809208747609178249385890097149096759852613655497818931297848216829989487226588048575640142704775551323796414515237462343645428584447952658678210511413547357395231134271661021359695362314429524849371871101457654035902799344037420073105785390621983874478084784896833214457138687519435064302184531910484810053706146806749192781911979399520614196634287544406437451237181921799983910159195618146751426912397489409071864942319615679452080951465502252316038819301420937621378559566389377870830390697920773467221825625996615014215030680384477345492026054146659252014974428507325186660021324340881907104863317346496514539057962685610055081066587969981635747363840525714591028970641401109712062804390397595156771577004203378699360072305587631763594218731251471205329281918261861258673215791984148488291644706095752706957220917567116722910981690915280173506712748583222871835209353965725121083579151369882091444210067510334671103141267111369908658516398315019701651511685171437657618351556508849099898599823873455283316355076479185358932261854896321329330898570642046752590709154814165498594616371802709819943099244889575712828905923233260972997120844335732654893823911932597463667305836041428138830320382490375898524374417029132765618093773444030707469211201913020330380197621101100449293215160842444859637669838952286847831235526582131449576857262433441893039686426243410773226978028073189154411010446823252716201052652272111660396665573092547110557853763466820653109896526918620564769312570586356620185581007293606598764861179104533488503461136576867532494416680396265797877185560845529654126654085306143444318586769751456614068007002378776591344017127494704205622305389945613140711270004078547332699390814546646458807972708266830634328587856983052358089330657574067

[illegible]

19966445547276083101187883891511493409393447500730258558147561908813987523578123313422798665035227253671712307568610450045489703600795698276263923441071465848957802414081584052295369374997106655948944592462866199635563506526234053 3...

$\dots f_{ddd=666} \simeq 1/1000$